

Lab 04:

Question#01:

CODE:

```
#include <iostream>
#include <string.h>
using namespace std;

class Employee
{
    char *EmployeeName;
    const int EmployeeId;

public:
    Employee(int id, const char *name = "N/A") : EmployeeId(id)
    {
        EmployeeName = new char[strlen(name) + 1]();
        strcpy(EmployeeName, name);
    }
    char *getName() { return EmployeeName; }
    void changeName(const char *name)
    {
        delete[] EmployeeName;
        EmployeeName = new char[strlen(name) + 1]();
        strcpy(EmployeeName, name);
    }
    int getID() const { return EmployeeId; }
    ~Employee() { delete[] EmployeeName; }
};

int main()
{
    Employee Employee1(1001, "Jhon"), Employee2(1002, "Jhonathan"),
    Employee3(1003, "Jhonny");
    cout << "IDs\tNames" << endl;
    cout << Employee1.getID() << "\t" << Employee1.getName() << endl;
    cout << Employee2.getID() << "\t" << Employee2.getName() << endl;
    cout << Employee3.getID() << "\t" << Employee3.getName() << endl;

    Employee2.changeName("Jhonny");

    cout << "\033[1;33m===After changing Employee2 name===\033[0m" << endl;
```

Object Oriented Programming C++

```
cout << "IDs\tNames" << endl;
cout << Employee1.getID() << "\t" << Employee1.getName() << endl;
cout << Employee2.getID() << "\t" << Employee2.getName() << endl;
cout << Employee3.getID() << "\t" << Employee3.getName() << endl;
return 0;
}
```

OUTPUT:

```
1001   Jhon
1002   Jhonathan
1003   Jhonny
====After changing Employee2 name====
IDs    Names
====After changing Employee2 name====
IDs    Names
1001   Jhon
1002   Jhonny
1003   Jhonny
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```

Question#02:

CODE:

```
#include <iostream>
using namespace std;

class DynamicArray {
private:
    int* arr;
    int size;

public:
    DynamicArray(int size) : size(size) {
        arr = new int[size];
        for (int i = 0; i < size; i++) {
            arr[i] = 0;
        }
    }
}
```

Object Oriented Programming C++

```
~DynamicArray() {
    delete[] arr;
}

void push(int value) {
    if (size > 0) {
        arr[size - 1] = value;
    }
}

int getSize() {
    return size;
}

void printArray() {
    for (int i = 0; i < size; i++) {
        cout << arr[i] << " ";
    }
    cout << endl;
}

};

int main(){
    DynamicArray myArray(5);
    myArray.push(10);

    cout << "Size of the array: " << myArray.getSize() << endl;
    cout << "Array elements: ";
    myArray.printArray();

    return 0;
}
```

OUTPUT:

```
{ .\CT-25276_Q2 }  
Size of the array: 5  
Array elements: 0 0 0 0 10  
PS D:\NED University\2nd Semester\Object Oriented Programming (OOPs)\Practicals\Lab 04> |
```

Question#03

CODE:

```
#include <iostream>  
using namespace std;  
  
class Account {  
private:  
    int account_no;  
    double account_bal;  
    int security_code;  
    static int object_count;  
  
public:  
  
    Account() {  
        account_no = 0;  
        account_bal = 0.0;  
        security_code = 0;  
        object_count++;  
    }  
  
    void initialize(int account_no, double account_bal, int security_code) {  
        this->account_no = account_no;  
        this->account_bal = account_bal;  
        this->security_code = security_code;  
    }  
  
    void print() {  
        cout << "Account Number: " << account_no << endl;  
        cout << "Account Balance: " << account_bal << endl;  
        cout << "Security Code: " << security_code << endl;  
    }  
};
```

Object Oriented Programming C++

```
    }

    static int getObjectCount() {
        return object_count;
    }
};

int Account::object_count = 0;

int main(){
    Account acc1;
    acc1.initialize(12345, 1000.0, 6789);
    cout << "Account " << Account::getObjectCount() << ":" << endl;
    acc1.print();
    cout << endl;

    Account acc2;
    acc2.initialize(67890, 2000.0, 1234);
    cout << "Account " << Account::getObjectCount() << ":" << endl;
    acc2.print();
    cout << endl;

    cout << "Total number of Account objects: " << Account::getObjectCount() <<
endl;

    return 0;
}
```

OUTPUT:

```
Account 1:
Account Number: 12345
Account Balance: 1000
Security Code: 6789

Account 2:
Account Number: 67890
Account Balance: 2000
Security Code: 1234

Total number of Account objects: 2
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```

Question#04:

CODE:

```
#include <iostream>
using namespace std;

class Student {
private:
    const int rollNumber;
    string name;
    int age;

public:
    Student(int r, string n, int a) : rollNumber(r) {
        name = n;
        age = a;
    }

    void display() const {
        cout << "Roll Number: " << rollNumber << endl;
        cout << "Name: " << name << endl;
        cout << "Age: " << age << endl;
    }

    void changeAge(int newAge) {
        age = newAge;
    }
};

int main() {
    Student s1(101, "Ali", 20);

    s1.display();

    s1.changeAge(21);
    cout << "\nAfter updating age:\n";
    s1.display();

    return 0;
}
```

OUTPUT:

```
Roll Number: 101  
Name: Ali  
Age: 20  
  
After updating age:  
Roll Number: 101  
Name: Ali  
Age: 21  
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```

Question#05:

CODE:

```
#include <iostream>  
using namespace std;  
  
class HotelMercato {  
private:  
    const double rent_per_day;  
    string customer_name;  
    int days_stayed;  
  
public:  
  
    HotelMercato(string customer_name, int days_stayed) : rent_per_day(1000.85),  
customer_name(customer_name), days_stayed(days_stayed) {}  
  
    double getPerDayRent() const {  
        return rent_per_day;  
    }  
  
    double calculateRent() {  
        if (days_stayed > 7) {  
            return (days_stayed - 1) * getPerDayRent();  
        } else {  
            return days_stayed * getPerDayRent();  
        }  
    }  
}
```

Object Oriented Programming C++

```
void display() {  
    cout << "Customer name: " << customer_name << endl;  
    cout << "Days stayed: " << days_stayed << endl;  
    cout << "Rent: " << calculateRent() << endl;  
}  
};  
  
int main(){  
    HotelMercato customer1("Faarid Amir", 20);  
    customer1.display();  
  
    return 0;  
}
```

OUTPUT:

```
Customer name: Faarid Amir  
Days stayed: 20  
Rent: 19016.2  
PS D:\NED University\2nd Semester\Object Oriented Programming (OOPs)\Practicals\Lab 04> |
```